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Charging flap system and motor vehicle

Abstract

A charging flap system for a motor vehicle that is at least partially operated electrically has a charging connection, which is arranged in a charging well, for electrically charging the motor vehicle, in particular a battery of the motor vehicle, a pivotable, outer covering flap which covers the charging well in a closed position and opens up same in an open position, and a pivotable, inner charging flap which opens up the charging connection in a charging position. The inner charging flap and the outer covering flap interact in such a manner that the outer covering flap and the inner charging flap are coupled to each other in a form-fitting manner in an intermediate position between the open position and the closed position.

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Background/Summary

BACKGROUND AND SUMMARY
(1) The invention relates to a charging flap system for an at least partially electrically operated motor vehicle, and to a motor vehicle.
(2) Charging flap systems for at least partially electrically operated vehicles are known from the prior art. A charging connector is received in a charging well which, in the motor vehicle, is disposed behind an access opening in a corresponding vehicle body component. In order for the access opening to be closed, an outer covering flap which, by way of a hinge arm, is adjustable between an opened and a closed position is usually provided.
(3) If the covering flap is open over a comparatively long period of time, as is the case in particular in partially electrically operated vehicles when the latter are being charged outdoors over a comparatively long period of time, snow caused by the weather can enter the opened system, the snow completely filling the charging well, which subsequently may lead to icing. The hinge arm may be blocked as a result, such that the covering flap can no longer be closed. The consequence is that the snow, or the ice, respectively, has to be tediously removed, which compromises comfort and costs time.
(4) It is therefore an object of the invention to provide a charging flap system in which the covering flap can be reliably closed independently of the weather conditions.
(5) This object is achieved according to the invention by a charging flap system for an at least partially electrically operated motor vehicle having a charging connector for electrically charging the motor vehicle, in particular a battery of the motor vehicle, disposed in a charging well. The charging flap system furthermore has a pivotable, outer covering flap which, in a closed position, covers and, in an open position, uncovers the charging well, and a pivotable inner charging flap which, in a charging position, uncovers the charging connector. The inner charging flap and the outer covering flap here interact such that the outer covering flap and the inner charging flap in an intermediate position between the open position and the closed position are coupled to one another in a form-fitting manner. In the intermediate position, the charging well is partially covered by the outer covering flap. The invasion of solid or liquid substances from the outside can be minimized as a result thereof. Because the outer covering flap in the intermediate position is set at an angle and does not project so far from the vehicle as in the open position, the risk of persons, for example pedestrians, colliding with the opened covering can be reduced.
(6) When closing the outer covering flap, the pressure applied causes the form-fitting coupling, the latter being in particular in the form of the two flaps latching or jamming together.
(7) The form-fitting coupling is preferably established by a releasable tongue-and-groove

connection, wherein the outer covering flap provides the tongue and the inner charging flap provides the groove, or vice versa.
(8) In one embodiment, the outer covering flap is assigned a receptacle region, in particular a receptacle cavity, in which at least part of the inner charging flap can be received. This enables the coupling of the two flaps.
(9) The receptacle is in particular understood to be the inner charging flap latching or jamming in the receptacle region of the outer covering flap.
(10) Optionally, a receptacle region can be provided on the inner flap and an appendage can be provided on the outer covering flap, the appendage penetrating the receptacle region of the inner flap when the two flaps are coupled.
(11) A further embodiment provides that the outer covering flap comprises a hinge arm by way of which the outer covering flap is attached to the vehicle, wherein the hinge arm has a receptacle region, in particular a receptacle concavity, in which at least part of the inner charging flap can be received. This enables the coupling of the two flaps.
(12) Optionally, the hinge arm can serve as a gate guide on which the inner charging flap slides along toward the receptacle region when the outer covering flap is being closed.
(13) Alternatively, an appendage can be provided on an end of the hinge arm proximal to the covering flap, and a receptacle region can be provided on the inner charging flap. When the outer covering flap is being closed, the receptacle region of the inner charging flap slides along the hinge arm toward the appendage, the latter then penetrating the receptacle region of the inner charging flap.
(14) According to a further embodiment, a coupling member by way of which the inner charging flap in the intermediate position is coupled in a form-fitting manner to the outer covering flap is provided on the inner charging flap. The inner charging flap can penetrate the receptacle region of the outer covering flap and be mounted, in particular latched or jammed, in a form-fitting manner by way of the coupling member.
(15) The coupling member can be integrally molded on the inner charging flap or, as a separate component, be fastened to the inner charging flap.
(16) The coupling member is preferably a protrusion which is molded on or fastened to the inner charging flap.
(17) The coupling member is in particular provided on a free end of the inner charging flap, in particular wherein the free end in terms of the inner charging flap is opposite the pivot axis of the inner charging flap. The free end comprises the region which when closing and opening the flaps comes into contact with the outer covering flap. It is therefore expedient for the coupling member to be disposed in this region.
(18) The free end is (substantially) the region of the inner charging flap that is the most remote from the pivot axis of the inner charging flap.
(19) The pivot axes of the two flaps preferably lie so as to be mutually parallel and in terms of the charging connector are disposed on the same side. In other words, the pivoting direction of the two flaps when opening and closing is (substantially) identical.
(20) Optionally, a latching cam can be provided on the coupling member. The latching cam guarantees the form-fitting coupling, in particular the form-fitting latching or jamming of the two flaps.
(21) For this purpose, the thickness of the latching cam (and the thickness of the coupling member) can be somewhat greater than the thickness of the receptacle opening of the receptacle region.
(22) The latching cam can interact with a latching well disposed in the receptacle region of the outer covering flap, so as to reinforce the form-fit of the coupling of the two flaps.
(23) One aspect provides that the outer covering flap in the intermediate position is held at an angle of at least 20° to at most 60°, in particular of at least 25° to at most 50°, preferably of at least 30° to at most 40°, in relation to a plane in which the charging connector is disposed, and/or that the inner

charging flap in the intermediate position of the outer covering flap is held at an angle of at least 90° to at most 180°, in particular to at most 150°, preferably to at most 120°, in relation to a plane in which the charging connector is disposed. A sufficient angular position of the outer covering flap is thus ensured, as a result of which the invasion of solid or liquid substances from the outside and the risk of people colliding on the outer covering flap can be reduced.
(24) The two angled flaps conjointly form a type of protective housing about a hinge box in which the hinge arm is pivotably received, so that—if at all—only a minor invasion of solid or liquid substances into the hinge box is possible.
(25) According to a further aspect, the form-fitting coupling of the outer covering flap and of the inner charging flap is releasable by a pivoting movement of the outer covering flap in the direction of the open position. The two flaps can thus be decoupled without tools, the latching or jamming thus being in particular cancelled without tools. Thereafter, the inner charging flap and the outer covering flap can be guided, in particular successively, to the respective closing positions thereof.
(26) According to a further aspect, the inner charging flap and/or the outer covering flap are/is preloaded to the respective charging or open position, respectively. In this way both flaps can be opened easily. Moreover, it can be prevented that the outer covering flap is closed without the inner charging flap having been closed beforehand, because the inner charging flap is interlocked with the receptacle region of the outer covering flap.
(27) It can also be provided that the outer covering flap is preloaded in the direction of the closed position thereof. The latching or jamming of the inner charging flap in the receptacle region of the outer covering flap here is not mandatory, because the outer covering flap is actively impinged toward the inner charging flap and the form-fit between the two flaps is thus actively maintained.
(28) The object is furthermore achieved by a motor vehicle having a vehicle body component and the charging flap system according to the invention.
(29) The advantages and properties described of the charging flap system according to the invention apply in analogous manner to the motor vehicle.
(30) Further advantages and properties of the invention are derived from the description hereunder and from the drawings to which reference is made.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic sectional view of a charging flap system according to an embodiment of the invention; and
(2) FIG. **2** is a view of the detail of the charging flap system according to FIG. **1**.
DETAILED DESCRIPTION OF THE DRAWINGS
(3) A charging flap system **10** for an at least partially electrically operated motor vehicle is illustrated in a schematic sectional view in FIG. **1**. The charging flap system **10** has a charging connector **12** which is disposed in a charging well **14**.
(4) An inner charging flap **16** which is pivotable about a pivot axis **17** is attached to the charging connector **12** by way of a fastening end **15**.
(5) Optionally, the inner charging flap **16** can also be provided in a region about the charging connector **12**.
(6) The inner charging flap **16** has a free end **19** which is (substantially) opposite the fastening end **15**. A coupling member **21**, which in the embodiment shown is configured as a molded protrusion, is provided on the free end **19**.
(7) Optionally, the coupling member **21** can also be configured as a separate component which is fastened to the free end **19** of the inner flap **16**.
(8) A hinge box **18** which, by way of a transition opening **20**, is connected to the charging well **14**

is disposed discretely from the charging well **14**.
(9) The charging well **14** and the hinge box **18** are partially enclosed by a housing **22** which is received in a vehicle body **24**.
(10) A hinge arm **26** to which an outer covering flap **28** is attached is pivotably mounted in the hinge box **18**. The hinge arm **26** on a side that faces away from the outer covering flap **28** has a receptacle region **30** which is configured as a receptacle concavity, for example as a groove or an opening.
(11) In a closed position (not illustrated) of the outer covering flap **28**, the charging well **14** is completely covered by the outer covering flap **28**. In other words, the housing **22** is closed in the closing position of the outer covering flap **28**. An external side **32** of the outer covering flap **28** here terminates so as to be flush with the surface of a body outer skin **34**.
(12) An angle α between the outer covering flap **28** and the body outer skin **34** in the closed position is accordingly (approximately) 0°.
(13) The inner charging flap **16** in a closed position (not illustrated) of the inner charging flap **16** bears in a planar manner on an internal housing of the charging connector **12** and hereby completely covers the charging connector **12**, or in particular closes the latter in the process.
(14) In the closed position, an angle β between the inner charging flap **16** and a fastening plane in which the inner charging flap **16** is fastened to the charging connector **12** is (approximately) 0°.
(15) In an open position (not illustrated), the outer covering flap **28** is pivoted away from the body outer skin **34**, as a result of which the charging well **14** is exposed. In other words, the housing **22** in the open position of the outer covering flap **28** is opened. The outer covering flap **28** here can project from the vehicle body **24** so as to be (approximately) orthogonal.
(16) The angle α between the outer covering flap **28** and the body outer skin **34** in the open position is in a range from 60° to 120°.
(17) The inner charging flap **16** in a charging position (not illustrated) of the inner charging flap **16** is pivoted away from the charging connector **12** and thereby exposes the latter.
(18) In the charging position, the angle β between the inner charging flap **16** and the fastening plane is more than 90°.
(19) In FIG. **1**, the outer covering flap **28** is illustrated in an intermediate position in which the outer covering flap **28** is coupled to the inner charging flap **16**. The coupling member **21** of the inner charging flap **16** here is received in a form-fitting manner in the receptacle region **30** of the outer covering flap **28** such that the outer covering flap **28** interlocks with the inner charging flap **16**.
(20) The angle α in the intermediate position is in a range from at least 20° to at most 60°, in particular from at least 25° to at most 50°, preferably from at least 30° to at most 40°.
(21) The angle β in the intermediate position is in a range from at least 90° to at most 180°, in particular from at least 90° to at most 150°, preferably from at least 90° to at most 120°.
(22) The receptacle region **30** and the coupling member **21** in the intermediate position of the flaps **16**, **28** are illustrated in detail in FIG. **2**.
(23) The outermost end of the coupling member **21** here has a wedge shape so as to guarantee easy sliding into the receptacle region **30**. Furthermore, a deformable latching cam **36** is provided on the coupling member **21** so as to reinforce the form-fit between the two flaps **16**, **28** and configure a type of jamming and/or latching.
(24) For this purpose, the coupling member **21** in the region of the latching cam **36** has a somewhat greater width B.sub.K than a width B.sub.A of the receptacle region **30** in the region of an opening, for example.
(25) A latching well **38** for receiving the latching cam **36** is disposed in the receptacle region **30**.
(26) In other words, the receptacle region **30** has an undercut.
(27) The shape of the receptacle region **30** is (substantially) complementary to that of the coupling member **21**.

(28) The sequence of a charging procedure while using the intermediate position of the outer covering flap **28** shown in FIGS. **1** and **2** will be described hereunder.
(29) A battery of the at least partially electrically operated motor vehicle, for example of a hybrid or electric vehicle, can be charged by way of the charging connector **12**. To this end, the outer covering flap **28** is opened and pivoted to the open position thereof.
(30) Thereafter, the inner charging flap **16** is opened so as to expose the charging connector **12**.
(31) Optionally, the inner charging flap **16** can be preloaded in the opening direction such that the inner charging flap **16** opens automatically upon opening the covering flap **28**.
(32) It can be provided that the inner charging flap **16** by opening the outer covering flap **28** is opened in a self-acting manner and, by virtue of the preload, transitions along the hinge arm **26** of the outer covering flap **28** to the open position thereof. The hinge arm **26** here serves as a type of guide.
(33) In the charging position, the inner charging flap **16** by way of the free end **19** thereof bears on the hinge arm **26** of the outer covering flap **28**.
(34) A charging plug is plugged into the exposed charging connector **12** and the battery of the motor vehicle is thus charged.
(35) The charging well **14** is completely exposed in the open position of the outer covering flap **28**. In this position, solid or liquid substances can invade the charging well **14** and the hinge box **18** unimpeded. The invading substances can accumulate in the hinge box **18** and lead to the hinge arm **26** being blocked such that the outer covering flap **28** can no longer be closed.
(36) Furthermore, the outer covering flap **28** in the open position projects from the vehicle body **24** so as to be (substantially) orthogonal and thereby extends far outward. In this position, there is the risk that vehicle users or pedestrians collide with the outer covering flap **28** and damage the outer covering flap **28**.
(37) In order to minimize or prevent the potential invasion of external substances and the risk of damage, the outer covering flap **28** is moved to the intermediate position.
(38) For this purpose, the outer covering flap **28** is pivoted from the open position in the direction of the closed position thereof. The coupling member **21** disposed on the free end **19** of the inner charging flap **16** here slides along the hinge arm **26** into the receptacle region **30** of the outer covering flap **28**.
(39) As a result of further pressure on the outer covering flap **28** in the direction of the closed position, the latching cam **36** bearing on the opening of the receptacle region **30** is deformed until the width B.sub.K of the coupling member **21** is somewhat smaller than the width B.sub.A of the opening of the receptacle region **30**. In other words, the latching cam **36** is compressed.
(40) The coupling member **21** thereafter slides further into the receptacle region **30** until the latching cam **36** is received in the latching well **38** and, as a result thereof, the coupling member **21** latches in a form-fitting manner to the receptacle region **30**.
(41) As a result of the interaction of the latching cam **36** and the latching well **38**, the outer covering flap **28** thus not only interlocks with the inner charging flap **16** but the two flaps **16**, **28** are coupled to one another in a form-fitting manner.
(42) Once the charging procedure has been completed, the form-fitting coupling of the two flaps **16**, **28** is released in that the outer covering flap **28** is pushed or pulled, respectively, in the direction of the open position thereof until the latching cam **36**—in a manner similar to the coupling procedure of the two flaps **16**, **28**—is deformed and the coupling member **21** can slide out of the receptacle region **30**.
(43) Subsequently, the inner charging flap **16** is pushed in to the closed position thereof, whereupon the outer covering flap **28** can be moved in to the closed position thereof unimpeded.

Claims

1. A motor vehicle, comprising: a vehicle body component; and a charging flap system for the motor vehicle, the charging flap system comprising: a charging well arranged in the vehicle body component; a charging connector arranged in the charging well, for electrically charging the motor vehicle; a pivotable, outer covering flap which covers the charging well in a closed position and uncovers the charging well in an open position; and a pivotable, inner charging flap which uncovers the charging connector in a charging position, wherein the inner charging flap and the outer covering flap interact such that the outer covering flap and the inner charging flap are coupled to one another in a form-fitting manner in an intermediate position between the open position and the closed position.
2. The motor vehicle according to claim 1, wherein the outer covering flap and the inner charging flap are coupled to one another in the form-fitting manner only in the intermediate position between the open position and the closed position.
3. The motor vehicle according to claim 1, wherein the form-fitting manner of the coupling is an interlocking coupling.
4. A charging flap system for an at least partially electrically operated motor vehicle, comprising: a charging connector arranged in a charging well, for electrically charging the motor vehicle; a pivotable, outer covering flap which covers the charging well in a closed position and uncovers the charging well in an open position; and a pivotable, inner charging flap which uncovers the charging connector in a charging position, wherein the inner charging flap and the outer covering flap interact such that the outer covering flap and the inner charging flap are coupled to one another in a form-fitting manner in an intermediate position between the open position and the closed position.
5. The charging flap system according to claim 4, wherein the outer covering flap comprises a hinge arm by way of which the outer covering flap is attached to the vehicle, and the hinge arm includes the receptacle region in which at least part of the inner charging flap is receivable.
6. The charging flap system according to claim 4, wherein the outer covering flap in the intermediate position is held at an angle (α) of at least 20° to at most 60° in relation to a plane in which the charging connector is disposed, and/or the inner charging flap in the intermediate position of the outer covering flap is held at an angle (β) of at least 90° to at most 180° in relation to a plane in which the charging connector is disposed.
7. The charging flap system according to claim 4, wherein the outer covering flap in the intermediate position is held at an angle (α) of at least 30° to at most 40° in relation to a plane in which the charging connector is disposed.
8. The charging flap system according to claim 4, wherein the inner charging flap in the intermediate position is held at an angle (β) of at least 90° to at most 120° in relation to a plane in which the charging connector is disposed.
9. The charging flap system according to claim 4, wherein the form-fitting coupling of the outer covering flap and the inner charging flap is releasable by a pivoting movement of the outer covering flap in the direction of the open position.
10. The charging flap system according to claim 4, wherein the inner charging flap and/or the outer covering flap is preloaded to the respective charging or open position, respectively.
11. The charging flap system according to claim 4, wherein the outer covering flap and the inner charging flap are coupled to one another in the form-fitting manner only in the intermediate position between the open position and the closed position.
12. The charging flap system according to claim 4, wherein the form-fitting manner of the coupling is an interlocking coupling.
13. The charging flap system according to claim 4, wherein the form-fitting manner of the coupling comprises a press-fit.
14. The charging flap system according to claim 4, wherein the outer covering flap has a receptacle region in which at least part of the inner charging flap is receivable.

15. The charging flap system according to claim 14, wherein the receptacle region comprises a concavity-shaped receptacle.
16. The charging flap system according to claim 4, wherein a coupling member, by way of which the inner charging flap in the intermediate position is coupled in the form-fitting manner to the outer covering flap, is provided on the inner charging flap.
17. The charging flap system according to claim 16, wherein a latching cam is provided on the coupling member.
18. The charging flap system according to claim 16, wherein the coupling member is provided on a free end of the inner charging flap.
19. The charging flap system according to claim 18, wherein the free end, in terms of the inner charging flap, is opposite a pivot axis of the inner charging flap.